THE EXTERIOR PRODUCT OVER A BASE OF DIMENSION ONE IS WELL DEFINED ON CHOW GROUPS IN DEGREE SHIFTED BY ONE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0FC4 (sources/stacks-chow/pinned-original-source.tex line 14658).

Allowed standard prerequisites (editorial).

- A normal scheme over a Dedekind domain is flat over it, and a point of a Dedekind spectrum is an effective Cartier divisor. More on Algebra, Lemma labelled more-algebra-lemma-dedekind-torsion-free-flat (Stacks tag 0AUW), invoked at proof lines 14686-14687. Supplies that the projection of a component dominating S is flat of the required relative dimension, which is what makes the flat pullback a bivariant class in the first case.
- Flat pullback, proper pushforward and the Gysin map of an effective Cartier divisor as bivariant classes. Same chapter, Lemmas labelled lemma-flat-pullback-bivariant (source line 6131), lemma-push-proper-bivariant (source line 6167) and lemma-gysin-bivariant (source line 6149), invoked at proof lines 14689-14692 and 14706-14715. Supply the three bivariant operations whose composites c' are capped against classes on Y to realise the cycle level product in each of the two cases.

Source reference. Original lemma 0AUW: sources/stacks-chow/tags.txt, source lines 13664–13664. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 14687 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0B78: sources/stacks-chow/pinned-original-source.tex, source lines 6131–6131. Same-chapter pointer into the author environment lemma at source line 6131; invoked at proof line 14690 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0EPK: sources/stacks-chow/pinned-original-source.tex, source lines 6167–6167. Same-chapter pointer into the author environment lemma at source line 6167; invoked at proof line 14692 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0B79: sources/stacks-chow/pinned-original-source.tex, source lines 6149–6149. Same-chapter pointer into the author environment lemma at source line 6149; invoked at proof line 14707 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *The map* $\times : \mathrm{CH}_n(X) \otimes_{\mathbf{Z}} \mathrm{CH}_m(Y) \to \mathrm{CH}_{n+m-1}(X \times_S Y)$ *is well defined.*

Proof. Consider n and m cycles $\alpha = \sum_{i \in I} n_i[X_i]$ and $\beta = \sum_{j \in J} m_j[Y_j]$ with $X_i \subset X$ and $Y_j \subset Y$ locally finite families of integral closed subschemes of δ-dimensions n and m. Let $K \subset I \times J$ be the set of pairs $(i, j) \in I \times J$ such that X_i or Y_j dominates an irreducible component of S. Then $\{X_i \times_S Y_j\}_{(i,j) \in K}$ is a locally finite collection of closed subschemes of $X \times_S Y$ of δ-dimension $n + m - 1$. This means we can indeed consider

$$\alpha \times \beta = \sum\nolimits_{(i,j) \in K} n_i m_j [X_i \times_S Y_j]_{n+m-1}$$

as a $(n + m - 1)$-cycle on $X \times_S Y$. In this way we obtain an additive map $\times : Z_n(X) \otimes_{\mathbf{Z}} Z_m(Y) \to Z_{n+m}(X \times_S Y)$. The problem is to show that this procedure is compatible with rational equivalence.

Let $i : X' \to X$ be the inclusion morphism of an integral closed subscheme of δ-dimension n which dominates an irreducible component of S. Then $p' : X' \to S$ is flat of relative dimension $n - 1$, see More on Algebra, Lemma 0AUW. Hence flat pullback along p' is an element $(p')^* \in A^{-n+1}(X' \to S)$ by Lemma 0B78 and hence $c' = i_* \circ (p')^* \in A^{-n+1}(X \to S)$ by Lemma 0EPK. This produces maps

$$c' \cap - : \mathrm{CH}_m(Y) \longrightarrow \mathrm{CH}_{m+n-1}(X \times_S Y)$$

which sends $[Y']$ to $[X' \times_S Y']_{n+m-1}$ for any integral closed subscheme $Y' \subset Y$ of δ-dimension m.

Let $i : X' \to X$ be the inclusion morphism of an integral closed subscheme of δ-dimension n such that the composition $X' \to X \to S$ factors through a closed point $s \in S$. Since s is a closed point of the spectrum of a Dedekind domain, we see that s is an effective Cartier divisor on S whose normal bundle is trivial. Denote $c \in A^1(s \to S)$ the Gysin homomorphism, see Lemma 0B79. The morphism $p' : X' \to s$ is flat of relative dimension n. Hence flat pullback along p' is an element $(p')^* \in A^{-n}(X' \to S)$ by Lemma 0B78. Thus

$$c' = i_* \circ (p')^* \circ c \in A^{-n}(X \to S)$$

by Lemma 0EPK. This produces maps

$$c' \cap - : \mathrm{CH}_m(Y) \longrightarrow \mathrm{CH}_{m+n-1}(X \times_S Y)$$

which for any integral closed subscheme $Y' \subset Y$ of δ-dimension m sends $[Y']$ to either $[X' \times_S Y']_{n+m-1}$ if Y' dominates an irreducible component of S or to 0 if not.

From the previous two paragraphs we conclude the construction $([X'], [Y']) \mapsto [X' \times_S Y']_{n+m-1}$ factors through rational equivalence in the second variable, i.e., gives a well defined map $Z_n(X) \otimes_{\mathbf{Z}} \mathrm{CH}_m(Y) \to \mathrm{CH}_{n+m-1}(X \times_S Y)$. By symmetry the same is true for the other variable and we conclude. □